

Kowndinya Boyalakuntla

Third-year Ph.D. student in Computer Science | Rutgers University

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Introduction

I am a third-year Ph.D. student in Computer Science at Rutgers University with nearly 4 years of experience in robot learning and planning. My research spans 3D scene understanding, object rearrangement, dynamic manipulation, and model-based reinforcement learning for humanoid locomotion. I have published at CoRL and IROS. My current focus is robust humanoid loco-manipulation using learned world models and vision-language-action (VLA) policies.

Education

Rutgers University Sep 2024 – present

Ph.D. in Computer Science; GPA: 3.85/4.0 Advisor: Prof. Abdeslam Boularias

Rutgers University Sep 2022 – Oct 2024

M.S. in Computer Science; GPA: 3.8/4.0 Advisor: Prof. Abdeslam Boularias

Thesis: Context-Aware Entity Grounding with Open-Vocabulary 3D Scene Graphs.

Indian Institute of Technology Tirupati Jul 2017 – Jul 2021

B.Tech. in Computer Science and Engineering; GPA: 8.42/10.0 Advisors: Prof. Sridhar Chimalakonda, Prof. Kalidas Yeturu

Research experience

Rutgers University Jan 2023 – present

Graduate Student Researcher · New Brunswick, NJ

Conducted research in robot learning for 3D scene understanding, object rearrangement, and placement.

RISHA Lab, IIT Tirupati Dec 2019 – May 2022

Undergraduate Student Researcher · Tirupati, India

Developed tools for GitHub repository evaluation and energy-aware mobile application generation.

University of Waterloo Jul 2020 – Sep 2020

Visiting Scholar · Waterloo, Canada

Improved RepoReaper's recall and reduced false positives in GitHub repository analysis without GHTorrent.

Publications

First-Author

1. **PL-MPC: Beyond Policy Alignment: Closing the Planning–Learning Loop for Robot Control with Learned World Models**

Kowndinya Boyalakuntla, Yuhan Liu, Abdeslam Boularias

Manuscript, 2026. Under review. [Paper] [Project] [Code]

- Combined hybrid multi-step value targets, uncertainty-aware MPC, and return-weighted policy distillation to improve planning and learning with a shared world model.
- Improved mean return over TD-M(PC)² from 98 to 387 on HumanoidBench balance-hard and from 199 to 466 on hurdle; demonstrated zero-shot transfer to KUKA wrench–nut alignment.

2. **Plan-Conditioned Imitation for Robust Object Retrieval under Self-Occlusion in Dense Clutter**

Kowndinya Boyalakuntla, Ajinkya Pawar, Abdeslam Boularias, Jingjin Yu

Manuscript, 2026. Under review. [Paper] [Project] [Code] [Data]

- Trained a recurrent policy with behavior cloning and DAGger to combine a fixed teacher plan with partial observations for object retrieval under self-occlusion.
- Achieved 90.7% success on 511 simulation scenes and 90% on UR5e; mean hardware task time was 67.3 s versus 192.7 s for the privileged teacher, which achieved 95% success.

3. **KARL: Kalman-Filter Assisted Reinforcement Learner for Dynamic Object Tracking and Grasping**

Kowndinya Boyalakuntla, Abdeslam Boularias, Jingjin Yu

IROS, 2025 [Paper] [Code]

- Integrated extended Kalman filtering for tracking recovery, a six-stage reinforcement learning curriculum, and grasp retries for dynamic grasping with an eye-in-hand camera.
- Achieved 93.3% versus 60.0% grasp success in regular scenes and 80.0% versus 53.3% in obstacle scenes against EARL on UR5e, with 15 trials per condition.

Co-Author

1. DAP: Diffusion-based Affordance Prediction for Multi-modality Storage

Haonan Chang, **Kowndinya Boyalakuntla**, Yuhua Liu, Xinyu Zhang, Liam Schramm, Abdeslam Boularias

IROS, 2024 [\[Paper\]](#) [\[Code\]](#)

- Collected data and evaluated a pipeline combining diffusion-based affordance sampling with point-cloud correspondence for 6D object placement.
- Achieved 98% book-shelving and 94% can-stacking success in simulation, versus RPDiff's 94% and 85%; demonstrated real-robot fruit storage after training on 80 demonstrations.

2. LGMCTS: Language-Guided Monte-Carlo Tree Search for Executable Semantic Object Rearrangement

Haonan Chang, Kai Gao, **Kowndinya Boyalakuntla**, Alex Lee, Baichuan Huang, Harish Udhaya Kumar, Jingjin Yu, Abdeslam Boularias

IROS, 2024 [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)

- Co-developed ELGR-Bench and implemented object selection, language parsing, and evaluation for a planner combining language-derived geometric priors with Monte Carlo tree search.
- Achieved 79.2% executable-plan success on ELGR-Bench versus 45.2% for LLMs as Few-Shot Planners and 74.1% for separate goal-pose and motion planning; demonstrated rearrangement on UR5e.

3. Context-Aware Entity Grounding with Open-Vocabulary 3D Scene Graphs

Haonan Chang, **Kowndinya Boyalakuntla**, Shiyang Lu, Siwei Cai, Eric Pu Jing, Shreesh Keskar, Shijie Geng, Adeeb Abbas, Lifeng Zhou, Kostas Bekris, Abdeslam Boularias

CoRL, 2023 [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)

- Built the DOVE-G benchmark (8 scenes, 4,000 queries) and evaluation pipeline for grounding language in open-vocabulary 3D scene graphs; reconstructed scenes and integrated ConceptFusion.
- Achieved 65.75% top-1 3D grounding success on DOVE-G versus OVIR-3D's 41.5%, using full queries and a 3D IoU threshold of 0.5.

Teaching experience

Fall 2026	CS440: Introduction to Artificial Intelligence, Course Instructor , Rutgers University
Spring 2026	CS440: Introduction to Artificial Intelligence, Teaching Assistant, Rutgers University
Fall 2025	CS460: Introduction to Computational Robotics, Teaching Assistant, Rutgers University
Spring 2025	CS440: Introduction to Artificial Intelligence, Teaching Assistant, Rutgers University
Fall 2024	CS440: Introduction to Artificial Intelligence, Teaching Assistant, Rutgers University

Peer review

- IEEE International Conference on Robotics and Automation (ICRA) (2026)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (2025, 2026)
- Conference on Robot Learning (CoRL) (2026)
- The International Journal of Robotics Research (IJRR) (2026)
- International Conference on Automated Planning and Scheduling (ICAPS) (2026)

Honors & awards

2022 Project showcase, IInvenTiv All IITs R&D Fair. Research project advised by Prof. Kalidas Yeturu, showcased at IIT Delhi.

2019 Gold Medal, Inter IIT Technical Meet 8.0. Ashoka's Tech for Change Challenge at IIT Roorkee; high-preparation event among 20 IITs.

Leadership & mentoring

Google Developer Student Clubs, IIT Tirupati

Feb 2019 – Jul 2020

Community Lead

Organized 12 workshops and hackathons for more than 70 students; led 15 mentoring sessions on Flutter and Android.

Professional service & activities

2021 Developed an Android application for IIT Tirupati's convocation.

2020 Completed 80 hours of National Service Scheme service; developed the IIT Tirupati NSS website and led its maintenance team.